

# The conjectured generating function of OEIS A162698, from the recurrence the entry states

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## Abstract

OEIS A162698 states a linear recurrence as a fact and carries a rational generating function as an empirical conjecture. The conjecture follows from the recurrence by polynomial algebra alone: the recurrence fixes the denominator, the published terms fix the numerator exactly, and the generating function is then determined as a rational function rather than checked on finitely many coefficients. No model of the sequence and no reading of its name is required.

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## 1 What the entry says

The first terms are 4, 5, 9, 11, 14, 17, 19, 23, ..., with offset 1. The entry states, as a fact,

$$a(n) = 2*a(n-1) - 2*a(n-2) + 2*a(n-3) - 2*a(n-4) + 2*a(n-5) - 2*a(n-6) + 2*a(n-7) - 2*a(n-8) + 2*a(n-9) - a(n-10) \text{ for } n > 10.$$

and separately records the conjecture

G.f.:  $x*(x^9+4*x^8-3*x^7+7*x^6-5*x^5+8*x^4-5*x^3+7*x^2-3*x+4) / ((x-1)^2*(x^4-x^3+x^2-x+1)*(x^4+x^3+x^2+x+1))$ . - \_Colin Barker\_, Dec 03 2012 ["Empirical" removed after Tom Edgar's comment by \_Andrey Zabolotskiy\_, Dec 15 2017]

The stated line was taken from the entry's factual formulas only. A formula qualified by a finite range – "holds at least up to  $n = 1000$  but is not known to hold in general" – is not a fact and is not used, nor is anything inside a conjectural block.

## 2 The argument

Write  $b_k = a(1+k)$  and

$$D(x) = 1 - \sum_i c_i x^i = x^{10} - 2x^9 + 2x^8 - 2x^7 + 2x^6 - 2x^5 + 2x^4 - 2x^3 + 2x^2 - 2x + 1,$$

the polynomial of the stated recurrence, of degree 10. Let  $B(x) = \sum_{k \geq 0} b_k x^k$ .

The recurrence says  $b_k = \sum_i c_i b_{k-i}$  for every  $k \geq 10$ . The coefficient of  $x^k$  in  $D(x)B(x)$  is exactly  $b_k - \sum_i c_i b_{k-i}$ , so it vanishes for every  $k \geq 10$  and

$$D(x)B(x) = P(x) = \sum_{k < 10} \left( b_k - \sum_i c_i b_{k-i} \right) x^k = x^{10} + 4x^9 - 3x^8 + 7x^7 - 5x^6 + 8x^5 - 5x^4 + 7x^3 - 3x^2 + 4x,$$

a polynomial whose every coefficient is fixed by finitely many published terms. Therefore

$$B(x) = \frac{P(x)}{D(x)} = \frac{x(x^9 + 4x^8 - 3x^7 + 7x^6 - 5x^5 + 8x^4 - 5x^3 + 7x^2 - 3x + 4)}{(x-1)^2(x^4 - x^3 + x^2 - x + 1)(x^4 + x^3 + x^2 + x + 1)},$$

which is the conjectured expression.

**Theorem 1.** *The conjectured generating function of OEIS A162698 is correct, given the recurrence the entry states.*

*Proof.* Immediate from the displayed identity:  $DB = P$  holds coefficientwise,  $D(0) = 1 \neq 0$  so  $D$  is invertible in the ring of formal power series, and  $B = P/D$  follows. This is an identity of rational functions, not a comparison of the 67 terms the entry publishes – those enter only in fixing the 10 coefficients of  $P$ .  $\square$